



TECHNICAL ANALYSIS IN FOREX MARKET

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0) Project objective & hypothesis

Goal: Identify short-term direction of CHF/JPY (H1, H4, D1) using EMA-50 and justify it with econometric evidence

TESTABLE HYPOTHESES:

- H0 (EMH/RWH): CHFJPY returns are unpredictable; technical signals have no predictive power.
- H1: Market shows short-run inefficiency/volatility clustering; EMA-50 signals help forecast next-period returns
- **DATA SOURCE AND SAMPLING:**

OANDA / METATRADER 5 API, SYMBOL: CHFJPY,
HORIZONS: H1&H4, SS: 50 CLOSED CANDLES
TIME CONVERSION TO UTC



1) Data pipeline (H1 & H4 CHFJPY)

Goal: Prepare clean, theory-consistent return data for econometric testing of market efficiency and predictability in CHF/JPY

Stationary

(Y_t is stationary if its statistical properties do not change over time)

$$E[Y_t] = \mu \quad Var(Y_t) = \sigma^2$$

$$Cov(Y_t, Y_{t-k}) = \gamma_k$$

(depends only on lag k , not on t)

vs

Non-Stationary

(statistical pattern changes over time, a random walk in prices)

$$\text{If } P_t = P_{t-1} + \varepsilon_t, \text{ then } Var(P_t) = t \cdot Var(\varepsilon_t)$$

which grows with time, so the variance is not constant.

This violates stationarity.

How We Make Data Stationary:

For a random-walk price: $P_t = \mu + P_{t-1} + \varepsilon_t$

$$r_t = \ln(P_t/P_{t-1}) = \ln(P_t) - \ln(P_{t-1}) = \varepsilon_t$$

$$E[Y_t] = \mu$$

$$Var(Y_t) = \sigma^2$$

$$Cov(Y_t, Y_{t-k}) = \gamma_k$$

non-stationary data like a moving escalator

$$E[r_t \mid I_{t-1}] = 0,$$

Efficient market(random walk)

$$E[r_t \mid I_{t-1}] \neq 0,$$

Predictable market

Include only candles whose time window has fully ended.

- A half-formed candle's close and volume are still changing $\rightarrow$ using it would create look-ahead bias, violating the econometric rule that all regressors must belong to the known information set I_{t-1} .



2) EMH foundation & Random Walk checks #1

Goal: Test whether CHF/JPY behaves as a random walk (efficient market) or shows a predictable structure.

Method: Apply ADF & KPSS tests on prices (P_t) and returns (r_t)

$$E[r_t \mid I_{t-1}] = 0,$$

Price changes(r_t) are random and independent → prices follow a random walk.

If I can reject this randomness, then trend-following (EMA, candlestick signals) becomes economically justified.

Efficient Market Hypothesis (EMH)

→

Weak form: Prices reflect all past market information.

→ Returns cannot be predicted using past prices or indicators.

→ Prices follow a random walk.

Model:

Hypotheses

ADF Test (Augmented
Dickey–Fuller)

$$\Delta P_t = \alpha + \rho P_{t-1} + \sum_{i=1}^p \gamma_i \Delta P_{t-i} + u_t$$

$H_0 : \rho = 0$ (*unit root, random walk*)

$H_1 : \rho < 0$ (*stationary*)

Fail to reject H_0

→

Random walk → efficient market

Prices non-stationary (unit root)

Reject H_0

→

Mean-reverting → predictability exists

Prices stationary



2) EMH foundation & Random Walk checks #2

Goal: Test whether CHF/JPY behaves as a random walk (efficient market) or shows a predictable structure.

Method: Apply ADF & KPSS tests on prices (P_t) and returns (r_t)

KPSS (Kwiatkowski–Phillips–Schmidt–Shin) :

KPSS Test Model :

$$P_t = \mu_t + \varepsilon_t$$

$$\mu_t = \mu_{t-1} + u_t$$

$$\varepsilon_t \text{ is stationary noise}$$

$$\mu_t \text{ is a random walk trend component}$$

H_0 : stationary (mean-reverting) ($Var(u_t) = 0$)

H_1 : series is non-stationary ($Var(u_t) > 0$)

Expected pattern for CHF/JPY

Series	ADF	KPSS	Interpretation
Price (P_t)	Fail to reject $H_0 \rightarrow$ unit root	Reject $H_0 \rightarrow$ non-stationary	Prices follow random walk $\rightarrow$ efficient
Return (r_t)	Reject $H_0 \rightarrow$ stationary	Fail to reject $H_0 \rightarrow$ stationary	Returns fluctuate around mean ≈ 0 $\rightarrow$ suitable for modeling

3) Inefficiency / Autocorrelation Evidence

Goal: To test whether CHF/JPY returns are serially correlated or correlated with their own past

- If there is correlation, it means past returns help predict future returns → violates the Efficient Market Hypothesis (EMH).
- If there's no correlation, returns are independent → market behaves efficiently (random walk).

Under the weak-form EMH:

$E[r_t \mid I_{t-1}] = 0$ and $Cov(r_t, r_{t-k}) = 0 \forall k \geq 1$

- The current return r_t should have no statistical link with any past return $r_{t-1}, r_{t-2}, \dots$
- If $Cov(r_t, r_{t-1}) = 0$, it's evidence that prices have memory — a potential inefficiency.

AR(1) on Returns : $r_t = \alpha + \phi r_{t-1} + u_t$ where,

α : average return (drift), ϕ : autocorrelation coefficient, u_t (new unpredictable shock), r_t : current return

Hypothesis		Interpretation
$H_0 : \phi = 0$	No serial correlation → returns independent	Efficient market (random walk)
$H_1 : \phi \neq 0$	Returns depend on past values	Market inefficiency
$\hat{\phi} > 0$: positive autocorrelation → momentum/trend-following behavior (returns keep their direction).		
$\hat{\phi} < 0$: negative autocorrelation — prices tend to reverse direction.		

$r_t = 0.0001 + 0.25 r_{t-1} + u_t$ coefficient $\phi=0.25$ is statistically significant ($p < 0.05$).

3) Inefficiency / Autocorrelation Evidence

Goal: To test whether CHF/JPY returns are serially correlated or correlated with their own past

$$r_t = 0.0001 + 0.25r_{t-1} + u_t, \quad \text{coefficient } \phi=0.25 \text{ is statistically significant (} p < 0.05 \text{).}$$

→ 25% of today's return is explained by yesterday's return, market shows short-term momentum → weak-form inefficiency → EMA is justified.

→ If instead $\phi \approx 0$ and insignificant → returns are random → EMH holds.

ACF and PACF of Returns

ACF (Autocorrelation Function): $\rho_k = \text{Corr}(r_t, r_{t-k})$

PACF (Partial Autocorrelation Function):

→ Shows direct correlation between r_t and r_{t-k} after removing the effects of intermediate lags.

If both ACF and PACF show all lags near 0 → returns are independent (random).

If one or more early lags (example; lag 1 or 2, 1h or 4 hourly time frames) are significant, that's autocorrelation evidence:

→ Positive lags → momentum (trend continues).

→ Negative lags → mean reversion (trend reverses).

Example: CHF/JPY → H1 returns show the following:

ACF(1) = +0.18 → mild positive correlation

ACF(2) ≈ 0

DW ≈ 1.6 AR(1): $\phi=0.17$, $p = 0.04$

→

Returns are positively correlated → short-term momentum → weak-form inefficiency
→ technical indicators like EMA-50 can capture this trend.

3) Inefficiency / Autocorrelation Evidence

Goal: A statistical reason to trust or distrust EMA signals

Econometric Result

Implication for EMA Strategy

$\phi \approx 0, DW \approx 2$

Market random $\rightarrow$ EMA useless

Ignore EMA-based trades

$\phi > 0, DW < 2$

Momentum exists $\rightarrow$ EMA-50 trend signals are valid

Trend-following works

$\phi < 0, DW > 2$

Mean reversion $\rightarrow$ use opposite signals (contrarian)

Fade EMA crossovers

For Example:

$\rightarrow$

$\phi = 0.17, p = 0.04 \rightarrow$ statistically significant momentum

$DW = 1.6 < 2 \rightarrow$ confirms positive serial correlation

EMA-50 below candlesticks $\rightarrow$ visually consistent upward trend

$\rightarrow$

The AR(1) and DW results justify why EMA-based strategy captures real short-term predictability, the market is not fully efficient

4) EMA 50 - Exponential Moving Average over 50 periods.

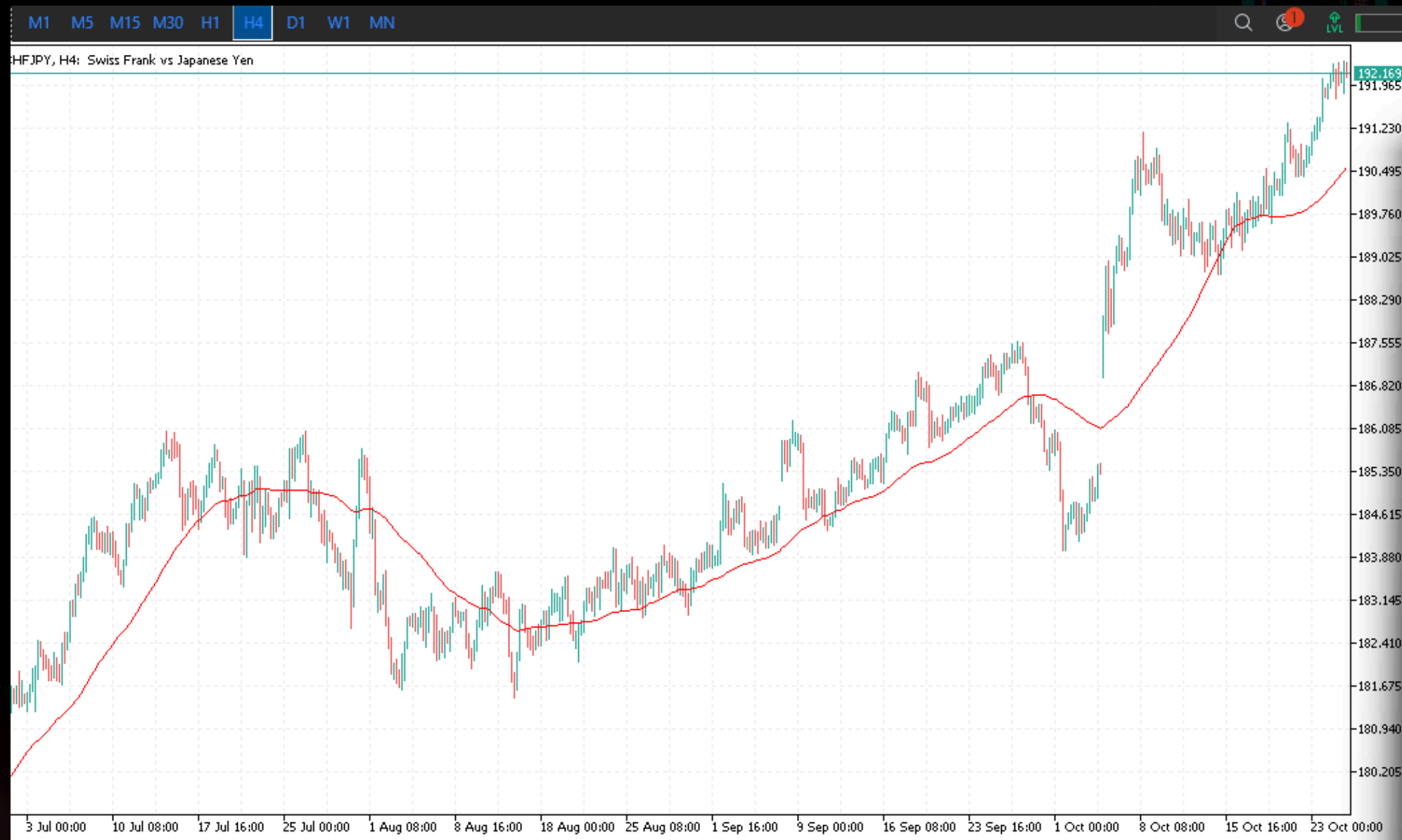
- It's a trend-smoothing tool that gives more weight to recent prices
- The "50" means capturing the medium-term market direction (roughly 2 trading months on a 4H chart or 50 hours on H1).

$$EMA_t = \lambda P_t + (1 - \lambda) EMA_{t-1}$$

$$\rightarrow \lambda = 2/(N + 1) \quad N=50 \rightarrow \lambda=2/51 \approx 0.039$$

EMA answers to the following question: "What is the market's average belief over the past 50 observations, giving more importance to the most recent ones?"

Practical and Economic Intuition



<http://bvi-en/cfds/platforms/metatrader-5/>

CHFJPY => 4-hourly chart

$$r_t = \alpha + \phi r_{t-1} + u_t$$

If $\phi > 0 \rightarrow$ past returns influence current ones $\rightarrow$ EMA curve trends upward smoothly.

If $\phi = 0 \rightarrow$ no relation $\rightarrow$ EMA flat and uninformative

EMA visually shows what AR(1) model measures statistically.

1. Price above rising EMA-50 Uptrend
Positive momentum — past returns predict future returns (weak-form inefficiency)
2. Price below falling EMA-50 Downtrend
Persistent negative momentum
3. Price crossing EMA-50 frequently No trend
Random walk — efficient market, no predictability

EMA Signal Alert Inbox x

 **billikhlasdiyor@gmail.com**
to me ▾

[GBPCHF - 4H] 2025-06-24 16:00:00

 Doji-like + EMA50 Signal

 EMA above candle

Price: 1.09943, EMA50: 1.10133

Distance: 0.00194

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[CHFJPY - 1H] 2025-06-24 08:00:00

 Doji-like + EMA50 Signal

 EMA above candle

Price: 179.12700, EMA50: 179.24641

Distance: 0.11991

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[EURCHF - 1H] 2025-06-24 10:00:00

 Doji-like + EMA50 Signal

 EMA below candle

Price: 0.94201, EMA50: 0.94060

Distance: 0.00139

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[USDCAD - 1H] 2025-06-24 11:00:00

 Doji-like + EMA50 Signal

 EMA above candle

Price: 1.37223, EMA50: 1.37332

Distance: 0.00109

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[GBPJPY.sml - 1H] 2025-06-24 09:00:00

 Doji-like + EMA50 Signal

 EMA below candle

Price: 197.21200, EMA50: 197.01211

Distance: 0.18639

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This automated Forex bot scans charts like EUR/USD and GBP/JPY using MetaTrader 5. It detects candlestick patterns (e.g., Doji, High-Wave)...

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DiyorTD/EMA-Project_Trading_BOT: This automated Forex bot scans charts like EUR/USD and GBP/JPY using...

This automated Forex bot scans charts like EUR/USD and GBP/JPY using MetaTrader 5. It detects candlestick patterns (e.g., Doji, High-Wave) with EMA-50 signals across 1H, 4H, and daily timeframes. R...

 GitHub

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Thank you

Q&A